माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर
MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR
Deemed to be University
(Declared under Distinct Category by Ministry of Education, Government of India)
NAAC ACCREDITED WITH A++ Grade
Gola Ka Mandir, Gwalior (M.P.)- 474005, INDIA
Ph.:+91-751-2409300, E-mail: vicechancellor@mitsgwalior.in, Website: www.mitsgwalior.in

Confidence-Aware RAG Pipeline with Hallucination Detection

Answer only when the sources prove it, and say why when they don't.

Shivam Gupta (BTCD24O1060)

Cornerstone Project · Project Expo 2026

Artificial Intelligence & Machine Learning Systems · July–Dec 2026

The problem: RAG still answers when it shouldn't



Out of scope

The documents don't cover the question, but the model answers anyway from its general training.



Invented details

The right passages are retrieved, but the model adds facts, numbers or causes they don't contain.



Contradiction

The model states something the passages actually say the opposite of.

The answer sounds equally confident every time. A user cannot tell a correct answer from a fabricated one, and a confident wrong answer does more damage than an honest "I don't know", because people act on it.

Objectives

1

Build
An end-to-end RAG pipeline on one laptop GPU with free, open models

2

Gate A
Check passage relevance before generating; refuse early if irrelevant

3

Gate B
Check every answer sentence against the passages after generating

4

Decide
Combine both into one confidence and answer, warn, or abstain with a reason

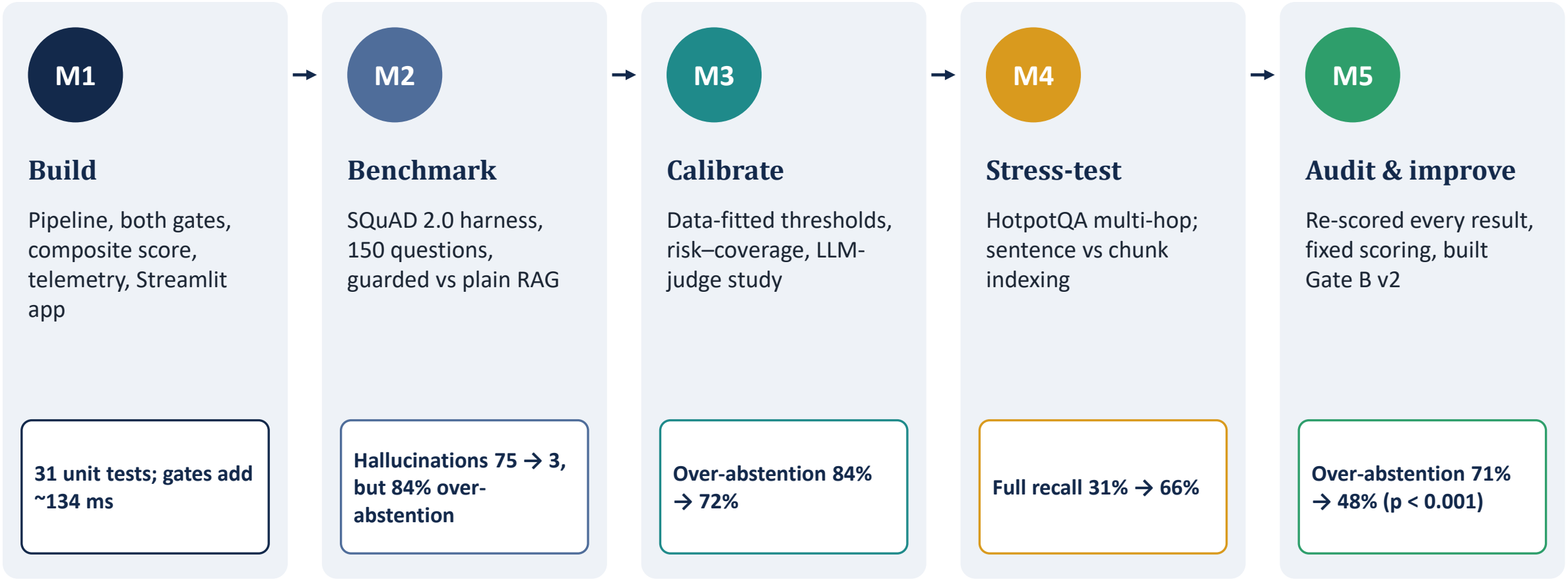
5

Monitor
Log every decision and show trends in a dashboard

6

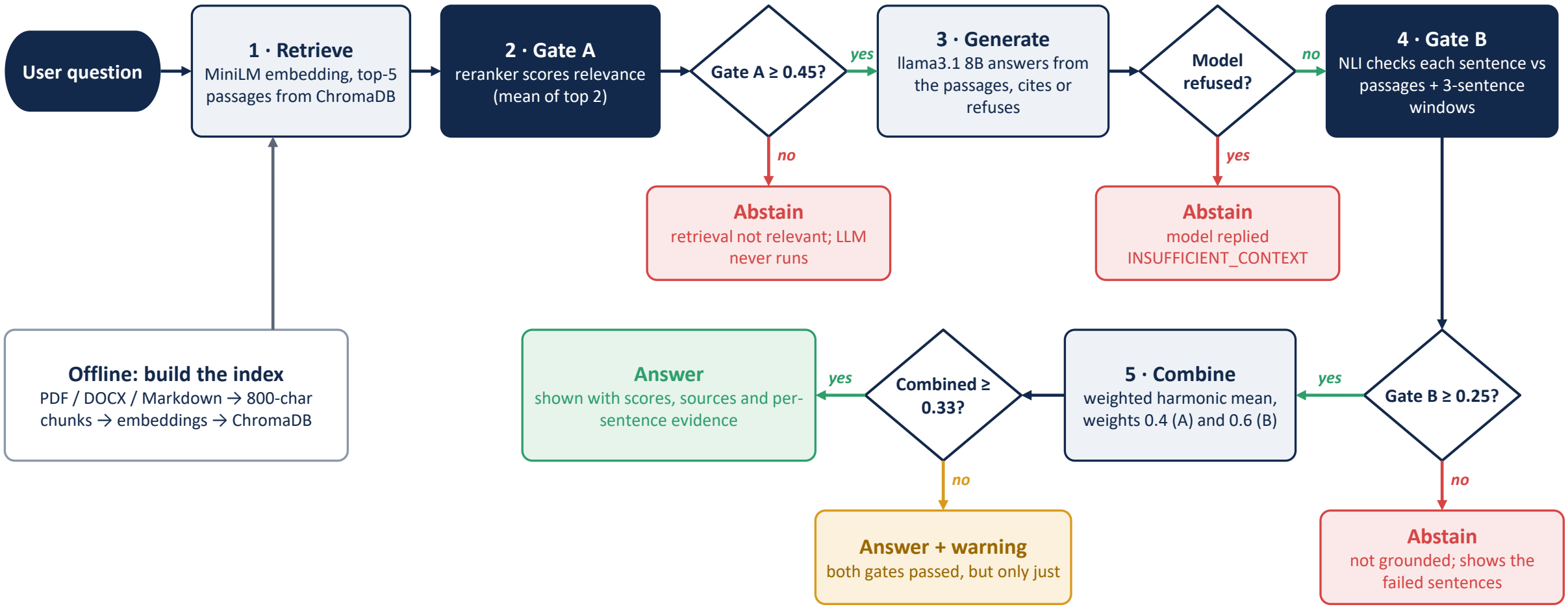
Measure
Compare honestly against plain RAG on SQuAD 2.0 and HotpotQA

Project workflow: five milestones



M5 went back over every saved result from M2–M4 before finalising the numbers in this talk.

System workflow: what happens to one question



Every outcome is logged to SQLite (scores, verdict, per-sentence labels, latency) and shown on the monitoring dashboard.

Three design decisions that matter

Two different models

- Before generating there is only a question, so relevance is a ranking problem: a cross-encoder reranker.
- After generating there is a statement to verify, so it is an entailment problem: an NLI model.
- Raw embedding similarity was rejected as a gate: its values aren't calibrated.

Harmonic mean, not average

Gate A = 0.95, Gate B = 0.05

Ordinary average

0.50 looks borderline: would pass

Harmonic mean

0.08 correctly near zero: fails

Good retrieval cannot buy credibility for an unsupported answer.

Contradiction veto

- If any sentence is judged a contradiction, Gate B is capped at 0.15.
- Nine true sentences cannot average away one false one.
- The cap is a fixed number, so recalibrating thresholds can never weaken it.

Milestone 1 · Building the pipeline

Layer	Files	Role
Data layer	ingest.py · embeddings.py · retriever.py	parse, chunk, embed, search
Scorers	scorers.py	reranker (Gate A), NLI (Gate B), LLM judge
Confidence layer	confidence.py	gates, contradiction cap, composite, verdicts
Generation	generator.py · llm.py	grounded prompt; Ollama, Groq, Gemini
Orchestration	pipeline.py · store.py	runs the five stages, logs one row per query
Interface	app.py · pages/	app, how-to-use guide, monitoring dashboard

Everything runs locally: no paid API, no data leaves the machine.

31 tests

model-free unit tests at M1 (92 today), running in under a second

16 ms

Gate A median latency on an RTX 4060

118 ms

Gate B median latency (v1); generation takes about 3 s

Milestone 2 · A fair benchmark

How each question is tested

- SQuAD 2.0: 150 questions, 75 answerable and 75 deliberately unanswerable
- Each indexed with its own paragraph plus 4 distractor paragraphs, in a fresh index
- Two arms, same model: the guarded pipeline vs a plain RAG baseline that always answers
- **Hallucination = answering an unanswerable question; over-abstention = refusing an answerable one**

75 → 3

unanswerable questions answered: plain RAG vs guarded (of 75)

84%

of answerable questions refused: far too cautious with placeholder thresholds

The gates work, but the thresholds were guesses.
Calibration was the next step.

Milestone 3 · Calibration and the judge study

Threshold	Before	Fitted	How
Gate A	0.45	0.45	Youden's J; AUC 0.65
Gate B	0.60	0.25	F1, kept above the 0.15 cap
Composite	0.55	0.33	just above both gates' corner

Gate A's threshold was stable: 0.45 was chosen again in 185 of 200 random half-splits of the data.

LLM-judge study (34 questions): the local NLI scorer and a gemma2 judge agreed no better than chance (κ ≈ 0.04), and the judge was ~860× slower. Neither is ground truth for the other.

84% → 72%

over-abstention after calibration

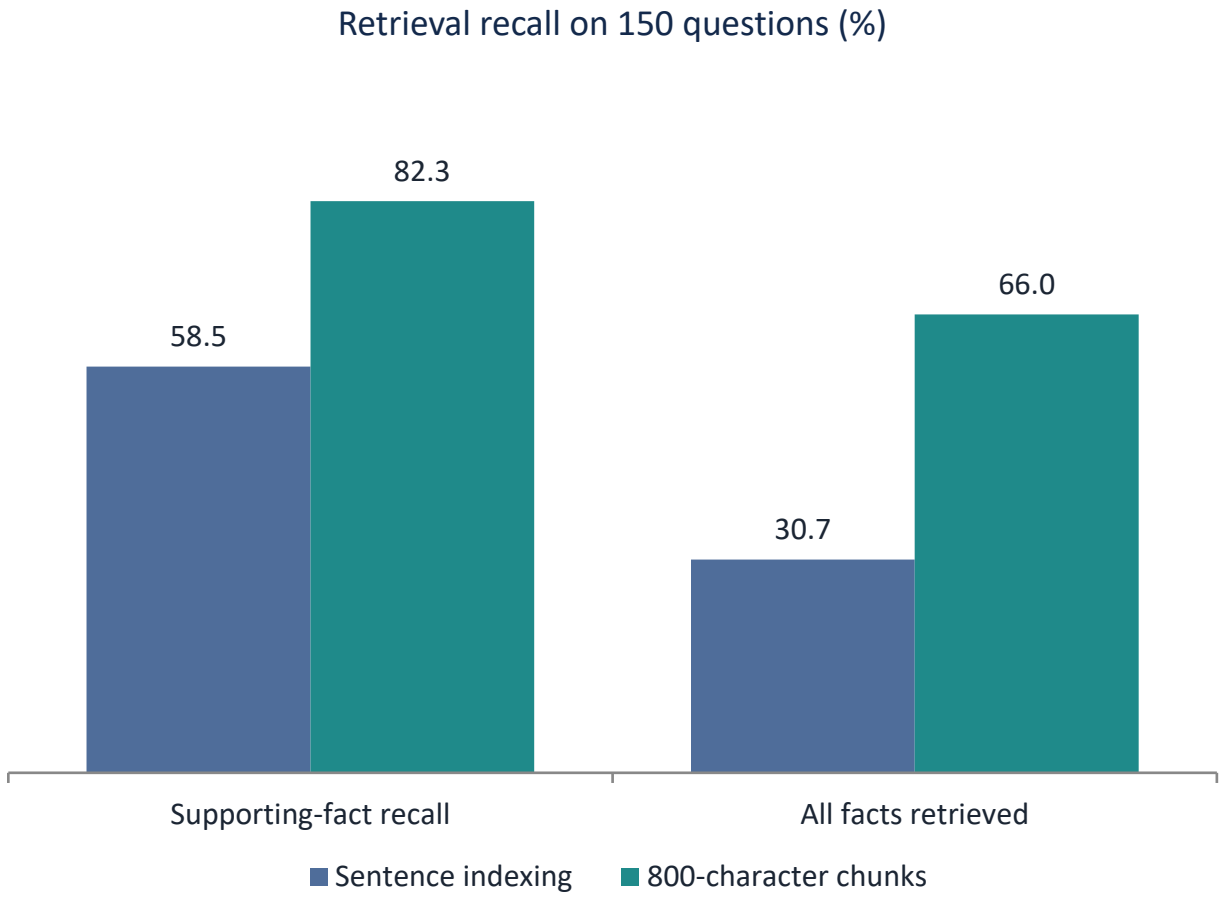
3 → 8

hallucinations of 75; paired test p = 0.06, not significant

AUC 0.65

Gate A separates answerable from unanswerable only modestly

Milestone 4 · Multi-hop questions (HotpotQA)



43% → 59%

plain RAG accuracy, sentence vs chunk indexing

41% → 24%

wrong answers among those given: gates on, but only 22% answered

~2 in 3

correct multi-hop answers rejected by Gate B: single-premise NLI can't chain evidence

Milestone 5 · Auditing every result

What the audit found	Why it mattered	Corrected picture
Citation scoring bug	"1894 [1]" scored as wrong, so every guarded answer got exact match 0	When it answered, it was right 20 of 21 times
Misleading headline	Aggregate exact match rewards abstaining on SQuAD 2.0	Report the answered/abstained breakdown instead
Prompt vs gates	The guarded prompt also lets the model refuse	Of 67 refusals: 40 by the gates, 27 by the model
Gate B misfiring	Gate B rejected 36 answerable answers	35 of those 36 were actually correct

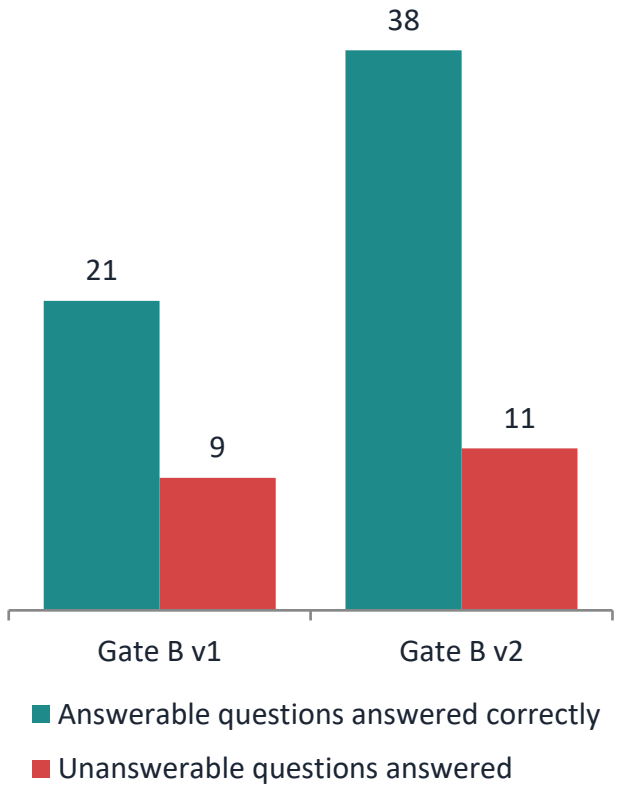
The system was more accurate than the earlier numbers showed, and the real bottleneck was Gate B, not the thresholds.

Milestone 5 · Gate B v2: the fix

Why Gate B failed

- The NLI model is trained on one- or two-sentence premises, but was given whole 100–350-word paragraphs
- It answered "neutral" even when one sentence stated the claim directly
- **v2: check each claim against the whole passage and against short 3-sentence windows; keep the best support**
- Windows alone failed on claims that summarise a list, so both are kept

Fresh runs, 75 + 75 questions



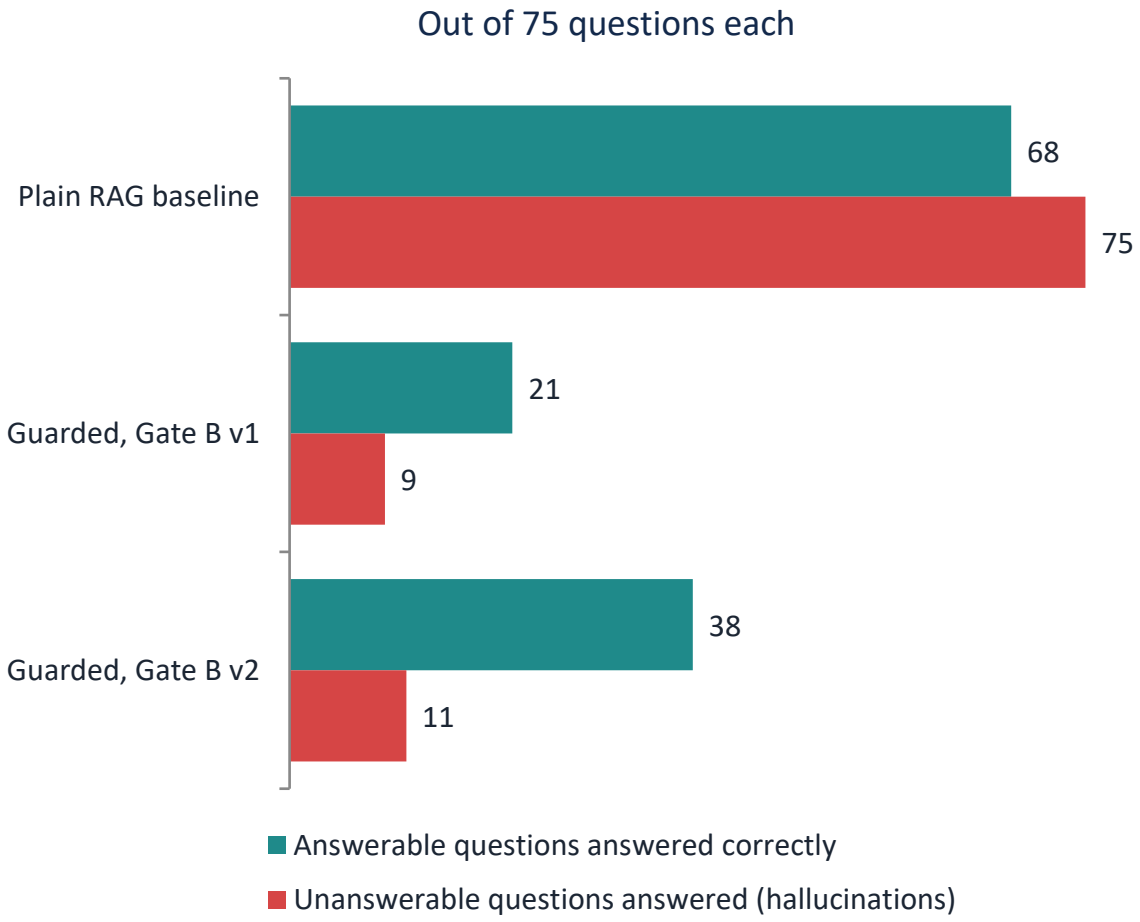
71% → 48%

over-abstention, $p < 0.001$ (17 gained, 0 lost)

+2

hallucinations of 75; $p = 0.5$, not significant

Final results on 150 SQuAD 2.0 questions



75 → 11

hallucinated answers (of 75): plain RAG vs guarded

97%

of its answers to answerable questions are correct (38 of 39)

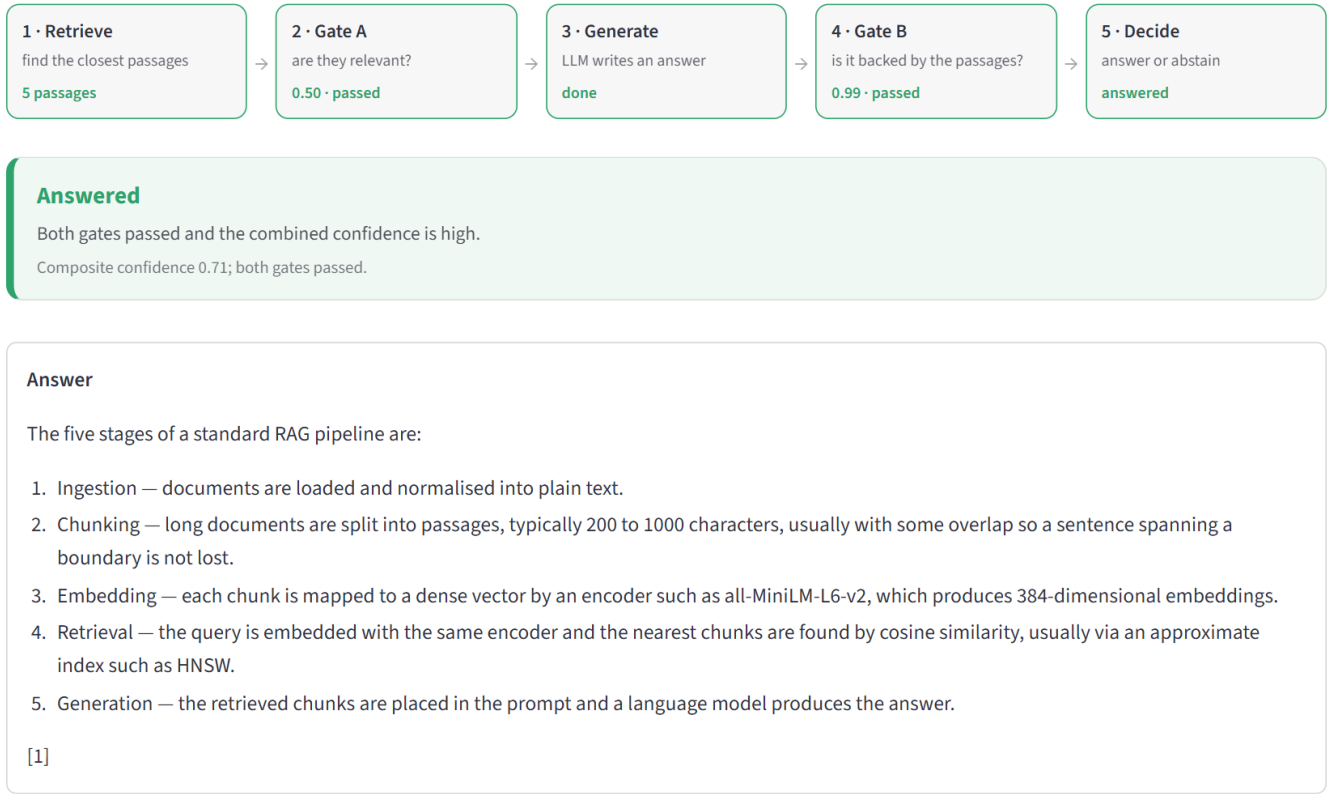
45% → 76%

of all answers given are correct: plain RAG vs guarded

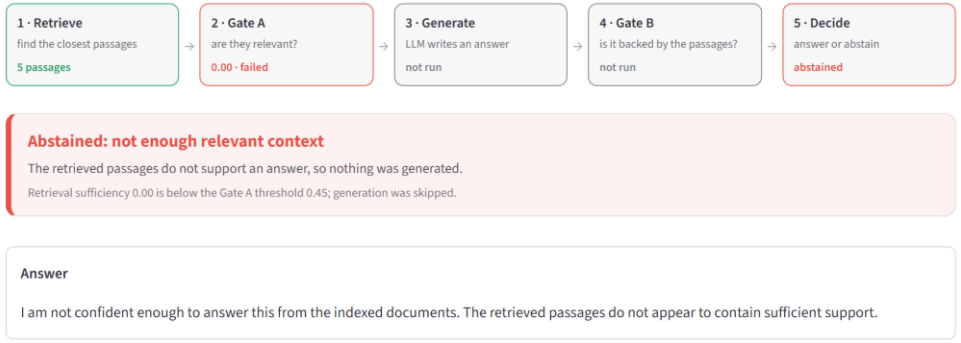
48%

of answerable questions still refused: the cost of caution

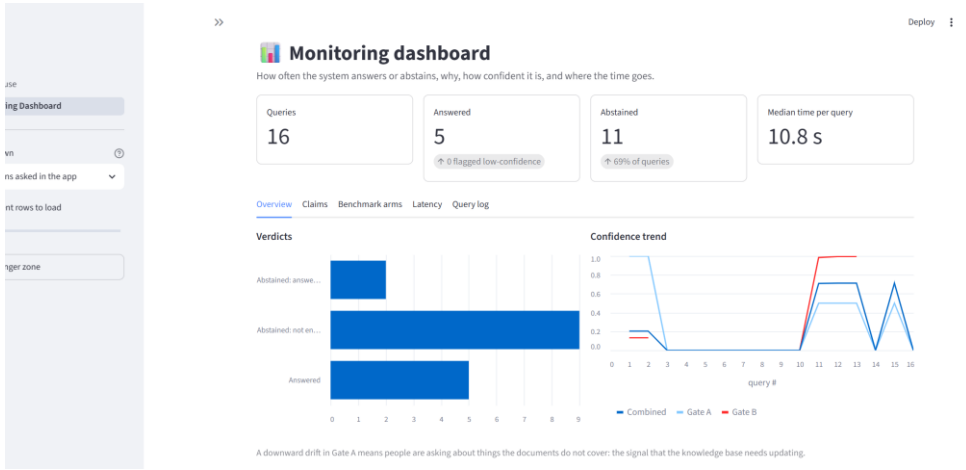
The app: every decision is visible



Answered: both gates pass, scores shown against their thresholds



Out of scope: Gate A refuses, the LLM never runs



Monitoring dashboard: verdicts, trends, latency

Limitations and future scope

Limitations

- Gate B still rejects some correct answers (all 18 of its final rejections were correct)
- Prompt and gates not fully separated: no prompt-only control run
- 150-question samples, so intervals are wide (hallucination 8–24%)
- Single-premise NLI can't verify multi-hop answers

Future scope

- **Larger NLI model: DeBERTa-large tested offline, AUROC 0.65 → 0.70 and false rejections 41% → 10% on SQuAD; needs more GPU memory**
- Prompt-only control arm to isolate the gates' effect
- 500+ questions with held-out calibration
- Multi-passage entailment or claim decomposition for multi-hop

Conclusion

1

Safer answers

Hallucinated answers fell from 75 to 11 of 75, and every refusal comes with its reason.

2

Measured honestly

Benchmarked against plain RAG, audited, and corrected where earlier numbers were wrong.

3

The next limit is clear

Gate B's entailment model, not the thresholds, decides how much coverage is lost.

Thank you · Questions?

Shivam Gupta (BTCD24O1060) · Cornerstone Project